

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (EC) Examination

May 2013

EC 310 Optical Communication

Date: 02.05.2013, Thursday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Explain the key elements of optical fiber communication system. [04]
(b) Compare step-index and graded index fiber. [03]
- Q - 2 (a) Enumerate the different attenuation mechanisms in fiber. Explain any one in detail. [05]
(b) Describe with a neat and labeled sketch the construction of edge emitting LED. [05]
(c) Explain three key transition processes involved in LASER action. [04]

OR

- Q - 2 (a) Explain Fabry Perot resonator cavity for LASER diode [05]
(b) Define and explain the significance of the following terms related to optical sources : [05]
(i) Radiance (ii) Charge carrier confinement (iii) Optical confinement
(c) A 30 Km long optical fiber has an attenuation of 0.4dB/km at 1310 nm. Find out the [04]
output optical power if 200 μ W of optical power is launched into the fiber.
- Q - 3 (a) Derive the expression for the power launched from the surface emitting LED source to [07]
step index and graded index fiber for radius of the source greater than and less than the
core radius.
(b) Describe operating principal of OTDR. With suitable diagram explain the cutback [07]
technique for attenuation measurement.

OR

- Q - 3 (a) Discuss the most common type of mechanical misalignment occurring between two [05]
joined fibers and its effect.
(b) Explain the setup for the eye diagram measurement. [05]
(c) Briefly explain equilibrium NA. [04]

SECTION – II

- Q - 4 (a) Define Following term: [03]
(i) Carrier multiplication (ii) Avalanche effect (iii) Dispersion in fiber
- (b) Explain quantum efficiency and responsivity of a photodiode. [04]
- Q - 5 (a) Compare silicon RAPD with PIN photo detector with respect to the structure, principle of operation and the performance. State the advantages of RAPD. [05]
- (b) Draw and explain schematic diagram of a typical optical receiver. [05]
- (c) A multimode silica fiber has a core refractive index $n_1=1.48$ and a cladding refractive index $n_2=1.46$. Calculate (i) numerical aperture and (ii) maximum acceptance angle for the fiber. [04]

OR

- Q - 5 (a) Explain principle of operation, the structure and the characteristics of the pin diode. [05]
- (b) Briefly discuss digital signal transmission in optical cable. [05]
- (c) A photodiode has a quantum efficiency of 65% when photons of energy $1.5 \times 10^{-19} \text{ J}$ are incident upon it. [04]
(i) At what wavelength is photo diode operating?
(ii) Calculate incident power required to obtain a photocurrent of $2.5 \mu\text{A}$ when the photodiode is operating as described above.

- Q - 6 (a) Discuss optical power loss model for point to point link with necessary diagram. [06]
- (b) Explain 2x2 fiber coupler and also discuss different losses associated with it [06]
- (c) What do you mean by population inversion? [02]

OR

- Q - 6 (a) Draw and explain basic structure of SONET Frame. [06]
- (b) Write a short note on Fiber Grating Filters [06]
- (c) What are the key system features of WDM? [02]
